Technische Hochschule Rosenheim

Speech and Language Processing!

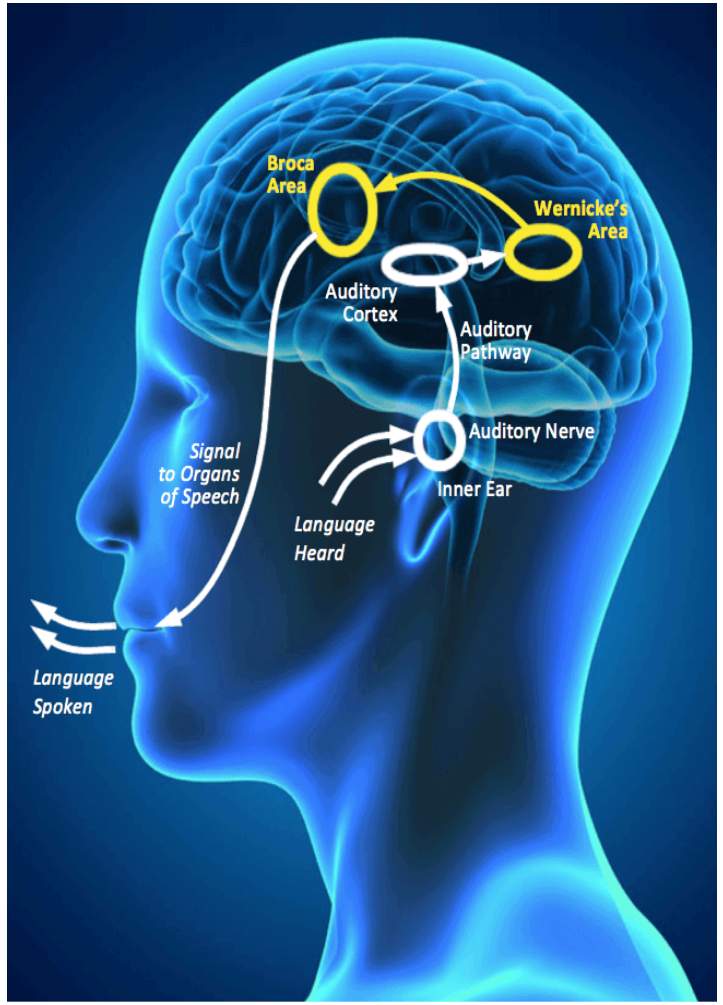
Technische Hochschule Rosenheim

On today's menu...

- Admin
- Why speech and language processing?
- History / Trivia
- Get to know each other
- Basics of machine learning

Why Speech and Language Processing?

- Speech is the most natural means of communication
- Active area of research
- Prerequisite for a wide range of applications
- Cross-domain, interdisciplinary
- Language processing is the present and the future
- Aside from technology, an understanding of speech and language is important



Who am I and why am I doing SLP?

- Sebastian Bayerl
  - Studied in Rosenheim
  - Originally from the region
  - Ph.D. in Erlangen/Nuremberg at Friedrich Alexander University
- Until recently:
  - Stand-in Professor for Applied AI at Rosenheim
  - before that: researcher in Nuremberg with industry placements
- Research Interests:
  - Speech processing
  - Paralinguistics
  - Pathological speech processing

Admin

- Course is Thursdays 08:00 - 11:15 in A0.03
- Course Repo: https://inf-git.th-rosenheim.de/base830/speech-and-language-processing-ws2627
- BYOD! + WFH!
- PStA (Experimental format!)

Who am I talking TO?

What's useful prior knowledge to make the best out of this course:

- Some probability theory
- Neural network knowledge
- Programming in Python
- Self-directed learning ability
- Perseverance
- A curious mind

Fiction

Speech Processing: History and Trivia

Star Trek (1966) and TNG (1987)

https://www.youtube.com/watch?v=tpKx7Oi0oeM



2001: A Space Odyssey (1968)

https://www.youtube.com/watch?v=ARJ8cAGm6JE



Star Wars (1977)

https://www.youtube.com/watch?v=EIZfE1AVDPQ



Knight Rider (1982)

https://www.youtube.com/watch?v=OY_nGtN4ARA



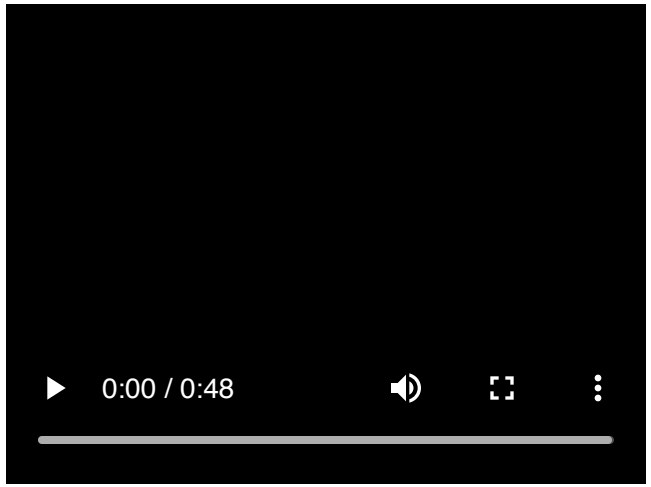
Charlie's Angels (2000)

https://www.youtube.com/watch?v=rRHRa80wVq8



Wild Hogs (2007)

https://www.youtube.com/watch?v=qVZtE3rL_sQ



Big Bang Theory (2009)



Her (2013)

https://www.youtube.com/watch?v=ne6p6MfLBxc



Reality

Speech Recognition: History and Trivia

Radio Rex (1920)

- "Classical" signal processing. "Recognition" of 500 Hz ("X")
- https://www.youtube.com/watch?v=AdUi_St-BdM





Bell Labs AUDREY (1952)

- Automatic digit recognition
- One speaker, isolated digits 0-9
- Based on acoustic / formant measurements

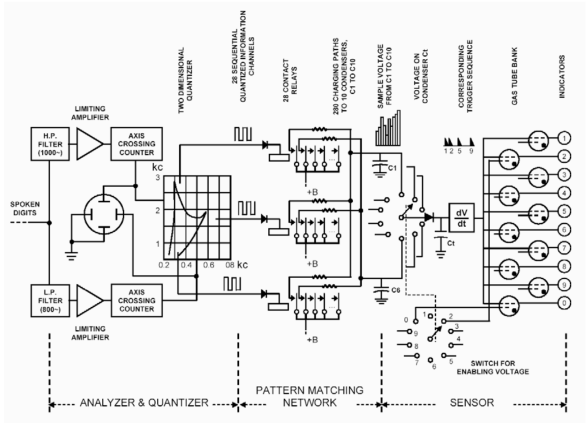


Figure 5 Block schematic of digit recognizer circuits. (Davis, Biddulph, and Balashek [9])

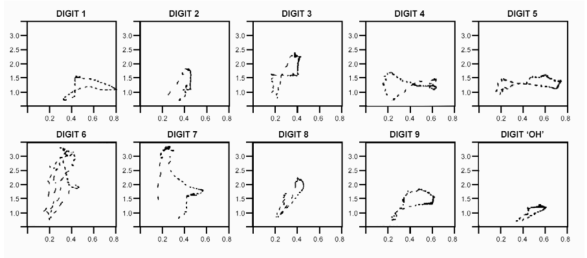


Figure 6 Photographs of formant 1 vs. formant 2 presentation of the digits. (Davis, Biddulph, and Balashek [9])

IBM Shoebox (1961/62)

- Recognizes digits 0-9 and six command words
- Voice-controlled arithmetic
- Demonstrated publicly at the 1962 Seattle World's Fair



https://mediacenter.ibm.com/media/(1961)%2BShoebox%2B-%2BIBM%2BArchives%2B(78-013)/0_4m2ynnkk

CMU HARPY (1976)

- Connected speech recognition
- Vocabulary of roughly 1,000 words
- Search over likely acoustic and syntactic paths (beam search)



https://www.bilibili.com/video/BV1iE411D7or?p=2

Speak & Spell (1978)

- Consumer speech synthesis
- LPC and early DSP technology
- Speech technology arrives as a children's toy



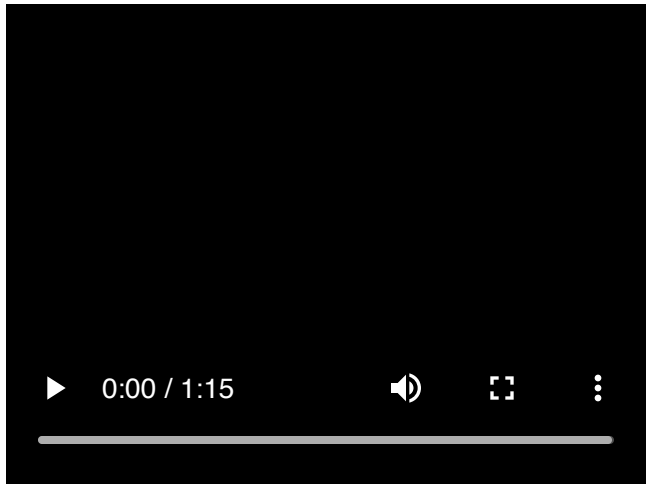
Worlds of Wonder's: Julie Doll (1987)

https://www.youtube.com/watch?v=UkU9Sblictc



Microsoft Speech Recognition (2008)

https://www.youtube.com/watch?v=-0kDcUEDfmY



BMW Speech Commands (2009)

https://youtu.be/NMS2VnDveP8



Siri (2011)



Microsoft Cortana (2015)

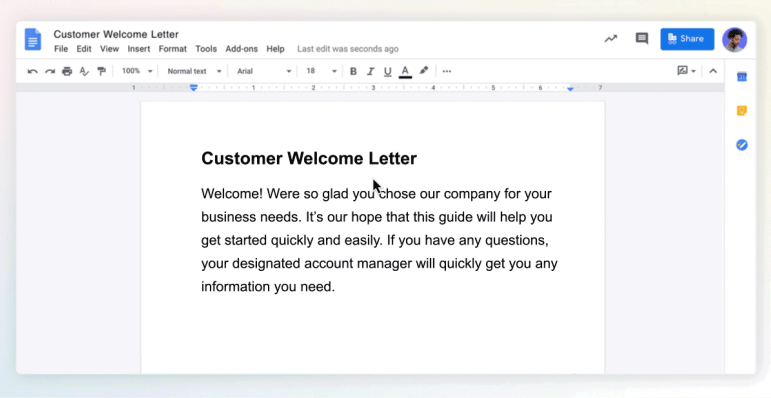
https://www.youtube.com/watch?v=96gPKLaSeHk



HELLO Barbie "She Talks" (2015)



Grammarly (since 2009 -)

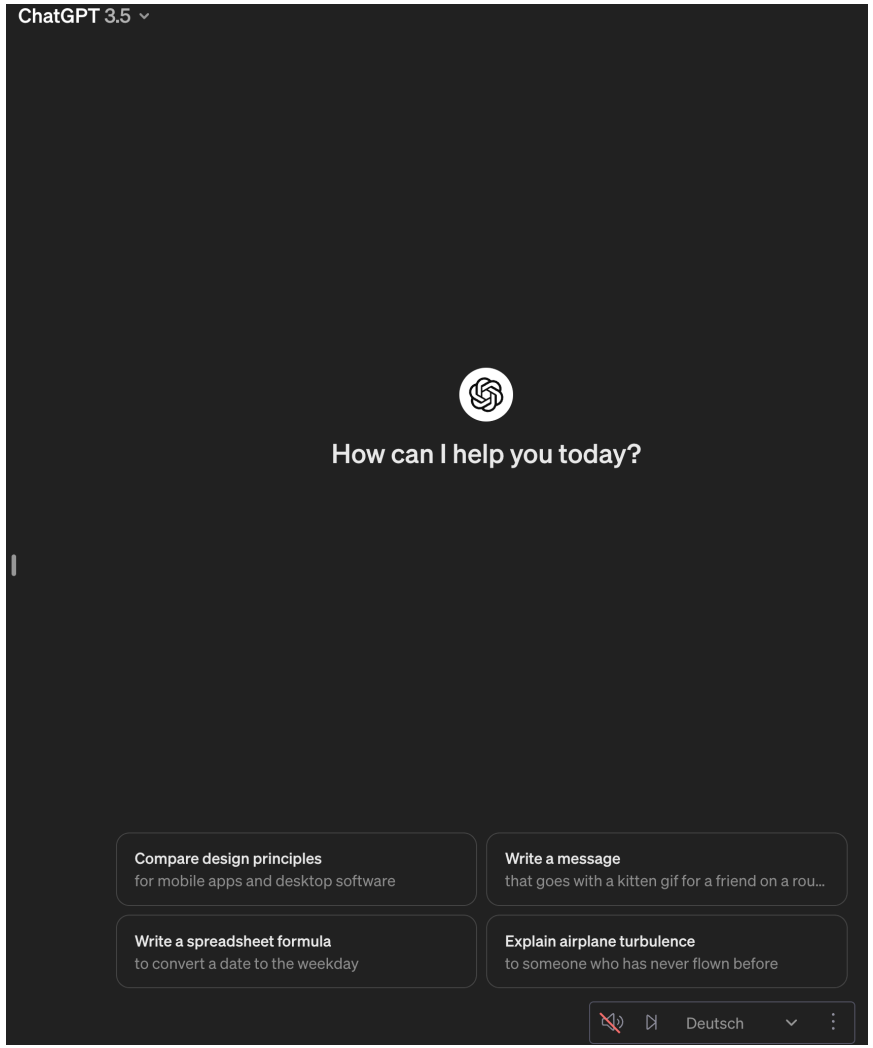


Google Neural Machine Translation (2016)



Image source: https://en.wikipedia.org/wiki/Google_Neural_Machine_Translation#/media/File:Ætoms_-_Translation.svg

Chat-GPT 2022/2023 ...





What about Research?

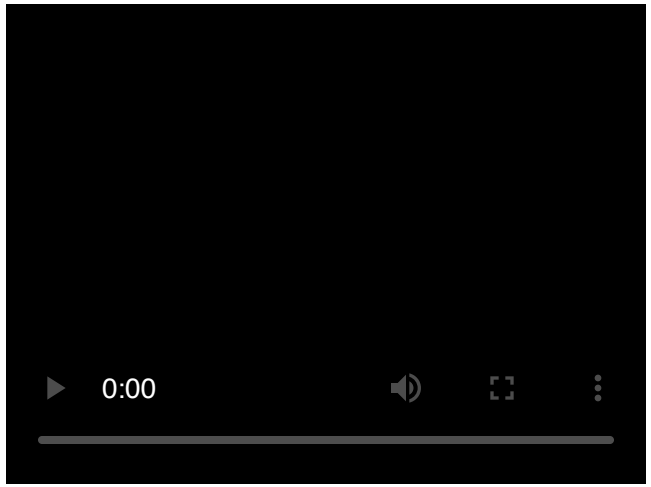
ELIZA (1966)

- https://psych.fullerton.edu/mbirnbaum/psych101/eliza.htm
- Joseph Weizenbaum (MIT)
- Imitates a psychotherapist
- Tries to "fake" human behaviour and fool humans (Turing Test)
- Structured lexicon
  - Looks for synonyms / generic terms for answers
  - e.g.: Mapping Mother / Family
    - User: I have problem with my mother
    - ELIZA: Tell me more about your family
- OOV: "tell me more" / "did not understand"



Verbmobil (1993-2000)

https://www.youtube.com/watch?v=DcG9-KWx0Fg



IARPA BABEL (2011-2016)

- 12 Languages
- Highly performant search without fixed keywords
- Needed around 10 hours of transcribed speech
- Highly scalable
  - ...world-wide surveillance of telecommunications service possible!



Rise of transformers (2018-)

- **BERT**: Bidirectional Encoder Representations from Transformers is a transformer-based machine learning technique for natural language processing (2018)
- **wav2vec 2.0**: A Framework for Self-Supervised Learning of Speech Representations
- Unsupervised Speech Recognition (wav2vec-U, 2021)





Rise of transformers and open models (2018-)

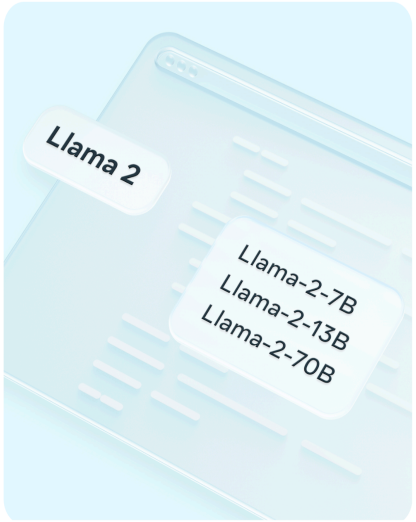
- WHISPER
- LLAMA

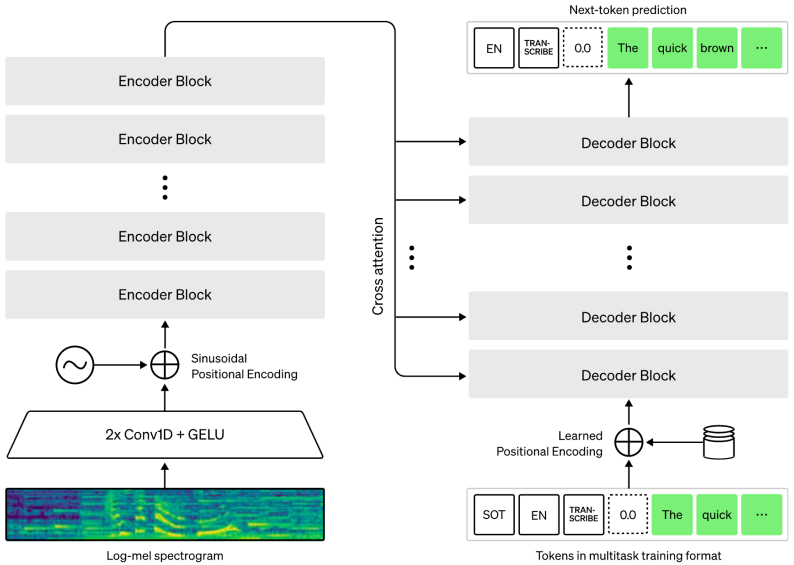
Large language model

Llama 2: open source, free for research and commercial use

We're unlocking the power of these large language models. Our latest version of Llama – Llama 2 – is now accessible to individuals, creators, researchers, and businesses so they can experiment, innovate, and scale their ideas responsibly.

Download the model





Current state of research

- Strong focus on research in large companies
  - META, Microsoft, Google, Amazon, OPENAI...
  - Especially Language Models and ASR need large amounts of resources
- Democratization of Speech processing
  - Kaldi
  - Hugging Face
  - ESPnet
  - Speech Brain







Top Research Topics in NLP

Recent Directions (2026)

Agentic and Long-Horizon NLP

LLMs acting in complex environments

- Agents plan, use heterogeneous tools, and interact over long trajectories
- Research shifts from simple tool calling to scheduling, recovery, adaptation, and efficiency
- Evaluation increasingly measures complete trajectories rather than isolated answers

Example papers

- AgencyBench: Benchmarking the Frontiers of Autonomous Agents in 1M-Token Real-World Contexts
  https://aclanthology.org/2026.acl-long.337/
- TPS-Bench: Evaluating AI Agents' Tool Planning & Scheduling Abilities in Compounding Tasks
  https://aclanthology.org/2026.acl-long.1614/

Reasoning and Test-Time Scaling

More computation at inference time

- Models spend additional inference compute on difficult problems
- Approaches include longer deliberation, multiple candidates, verification, and search
- Research increasingly treats the complete inference procedure as the system being evaluated

Example papers

- Test-Time Scaling in Reasoning LLMs: Inference Regimes, Evaluation, and Reproducibility
  https://aclanthology.org/2026.acl-long.1544/
- Ranking Reasoning LLMs under Test-Time Scaling
  https://arxiv.org/abs/2608.04001

Memory, Retrieval, and Context Management

Beyond simply increasing the context window

- Long-running systems need to decide what to retain, retrieve, compress, and forget
- Explicit working and persistent memory complement model context
- Retrieval is increasingly one component of larger agent memory architectures

Example papers

- Memory-R1: Enhancing Large Language Model Agents to Manage and Utilize Memories via Reinforcement Learning
  https://aclanthology.org/2026.acl-long.583/
- PACE: Predictive Adaptive Context Extraction for Long-Horizon LLM Agents
  https://aclanthology.org/2026.acl-long.1252/

Multilingual and Global NLP

From language coverage to actual capability

- Large models cover many languages, but capability remains uneven
- Research increasingly evaluates reasoning, cultural knowledge, and functional performance
- Benchmark translation itself can introduce substantial evaluation errors

Example papers

- MuBench: Assessment of Multilingual Capabilities of Large Language Models Across 61 Languages https://aclanthology.org/2026.findings-acl.794/
- GaoYao Benchmark: Evaluating Multilingual and Multicultural Abilities of Large Language Models https://arxiv.org/abs/2605.04045

Trustworthy and Reliable NLP

Reliability beyond hallucination

- Evaluation includes uncertainty handling, tool failures, policy compliance, and error recovery
- Agentic systems introduce cascading errors across long action sequences
- Research examines whether models know when information or capabilities are missing

Example papers

- CAR-bench: Evaluating the Consistency and Limit-Awareness of LLM Agents under Real-World Uncertainty
  https://aclanthology.org/2026.acl-long.1886/
- Aligning Agents via Planning: A Benchmark for Trajectory-Level Reward Modeling
  https://arxiv.org/abs/2603.07379

Multimodal Language Models

Language models increasingly see, hear, and act

- Text is becoming one modality within larger foundation models
- Joint reasoning over speech, images, video, and text is increasingly common
- Multimodal models connect NLP with speech, perception, and interactive agents

Example papers

- Audio-Visual Intelligence in Large Foundation Models
  https://arxiv.org/abs/2605.04045
- HEAR: Real Voices, Real Bias
  https://arxiv.org/abs/2601.20510

Top Research Topics in Speech Processing

Recent Directions in Speech and Spoken Interaction (2026)

Overview of major research directions in speech recognition, speech-language modeling, and spoken interaction

Speech and Audio Language Models

Beyond ASR + LLM pipelines

- Models directly reason over speech and other audio signals
- Speech representations include semantics, speaker information, prosody, and emotion
- Evaluation increasingly tests spoken-language understanding rather than transcription alone

Example papers

- MMSU: A Massive Multi-task Spoken Language Understanding and Reasoning Benchmark
  https://arxiv.org/abs/2506.04779
- HEAR: Real Voices, Real Bias
  https://arxiv.org/abs/2609.35952

Full-Duplex Spoken Dialogue

Listening while speaking

- Systems must handle interruptions, overlap, backchannels, and dynamic turn-taking
- Low latency becomes part of a broader conversational interaction problem
- Evaluation moves beyond WER toward timing, behavior, coherence, and interaction quality

Example papers

- Full-Duplex Interaction in Spoken Dialogue Systems: A Comprehensive Study from the ICASSP 2026 HumDial Challenge
  https://arxiv.org/abs/2604.21406
- AdaptDuplex: From Static to Adaptive Full-Duplex Spoken Dialogue
  https://arxiv.org/abs/2609.29217

Massively Multilingual and Low-Resource Speech

From hundreds toward global language coverage

- Foundation models support increasingly large language inventories
- The difficult problem is useful performance for low-resource and under-documented languages
- Research focuses on transfer, adaptation, data scarcity, and linguistic diversity

Example papers

- Omnilingual ASR: Speech Recognition for 1600+ Languages
  https://arxiv.org/abs/2511.09690
- Breaking the Transcription Bottleneck: Fine-tuning ASR Models for Extremely Low-Resource Fieldwork Languages
  https://arxiv.org/abs/2509.14128

Inclusive and Robust Speech Processing

Speech technology for diverse speakers

- Standard ASR still performs unevenly across disabilities, accents, dialects, and speaking conditions
- Research combines adaptation, augmentation, speaker information, and better datasets
- Accessibility is increasingly treated as a central evaluation problem

Example papers

- Speech Accessibility Project Challenge 2025
  https://arxiv.org/abs/2507.22047
- GLoRIA: Gated Low-Rank Interpretable Adaptation for Dialectal ASR
  https://arxiv.org/abs/2603.02464

Speech Security and Authenticity

When realistic synthetic voices become a problem

- Modern speech generation makes spoofing and impersonation increasingly difficult to detect
- Detectors must generalize to previously unseen generators
- Authentication, provenance, privacy, and misuse become central speech research problems

Example papers

- Audio Deepfake Detection in the Age of Advanced Text-to-Speech Models
  https://arxiv.org/abs/2601.20510
- A General Model for Deepfake Speech Detection: Diverse Bonafide Resources or Diverse AI-Based Generators
  https://arxiv.org/abs/2603.02464

Sources

https://soniox.com/wiki/history-of-speech-recognition

Late 2025 slide	October 2026	Change
Agentic NLP	**Agentic and Long-Horizon NLP**	Keep, but emphasize long trajectories, tool scheduling, recovery, cost and realistic evaluation.
Retrieval Grounded Generation	**Memory, Retrieval, and Context Management**	RAG is still important, but no longer deserves isolation as the frontier. Retrieval is becoming part of agent memory and reasoning loops. (arXiv)
Long Context Language Models	merged into **Memory, Retrieval, and Context Management**	Shift from “how many tokens fit?” to “what information should remain available over long tasks?”
no separate slide	**Reasoning and Test-Time Scaling**	**New major topic.** Inference-time compute, candidate sampling, search and verification became a distinct research area. (arXiv)
Multilingual and Global NLP	**Multilingual and Global NLP**	Keep. But change emphasis from number of languages to quality, cultural competence, cross-lingual transfer and benchmark validity. (ACL Anthology)

Late 2025 slide	October 2026	Change
Trustworthy NLP: hallucination, bias, evaluation	**Trustworthy and Reliable NLP**	Broaden considerably. For agents, reliability now includes tool availability, uncertainty, error propagation, policy compliance and entire trajectories. (ACL Anthology)
no separate slide	**Multimodal Language Models**	Add. NLP is increasingly embedded in systems jointly processing speech, vision, video and text. (arXiv)
Speech Language Models: LLM-based ASR	**Speech and Audio Language Models**	Broaden. The interesting problem is no longer only using LLMs to improve transcription, but direct speech understanding and reasoning. (arXiv)
Massively Multilingual ASR	**Massively Multilingual and Low-Resource Speech**	Keep. The novelty is moving from headline language counts toward useful performance for low-resource languages. Interspeech 2026 also foregrounded linguistic diversity. (Interspeech 2026)

Late 2025 slide	October 2026	Change
Streaming and Low Latency ASR	**Full-Duplex Spoken Dialogue**	**Major reframing.** Streaming recognition is now an enabling technology. The research frontier is interruption, overlap, listening while speaking, timing and conversational control. (arXiv)
Inclusive and Robust ASR	**Inclusive and Robust Speech Processing**	Keep. Broaden beyond WER to diverse speakers, disabilities, dialects, hearing/speech science and real-world conditions. (arXiv)
Efficient Adaptation in ASR	no standalone slide	**Demote.** LoRA, experts and PEFT remain active techniques, but they are now implementation mechanisms within multilingual, personalized and efficient speech systems rather than a top-level research direction. (arXiv)
no separate slide	**Speech Security and Authenticity**	**Add.** High-quality generative speech has made spoofing, impersonation and detector generalization a much more visible research problem. Interspeech 2026 devoted a keynote to countermeasures against misuse of speech generative AI. (arXiv)